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thickened the air in the pit, and then it would appear again; they lighted their candles at it sometimes when they went out; and fo in this pit it did no further human?

Damps.

In another pits, however, it from apparently, and at half produced a multi terrible explolions. "This was oca- fioned by none of the workmen going independent down with a lighted candle, after a ceilation of work for form days, and the force exerted by it femen equal to that of gun powder.

The formation of unintractable air in mines cannot be attributed to any variation of atmospheric or there is no natural process with which we are ac- quanted, by which fish a change could be accom- plished. In one influence, however, we have an ex- ample of a champ being produced, not only with- out any confidential flagation of atmospheric air, has where there is the belt circularism recognizable. This is in large bellows ufeid in metallurgie works, which are foretime birth by an explosion of infam- make matter processing from the mind matters with which the leather is greeded. Dr Priefley has flown, that inflammable air is comprised of pure ele- mentary fire charcoal or phologines and stable veter; and that this composition may take place even _in_ _wason._ All the materials are to be met with in the bowels of the earth. Coal, a bitumen imbalance, is abundantly able to supply the phogililon; the na- tural motions of the earth affords water, and the heat, between goods, which constantly calls in the bowels of the earth, may be sufficient to produce a quantity of fullammable air, which gradually occurring- falling in chois places where it is not a constant threat of atmospheric air to carry it off, will foam produce the desirable effects already mentioned.

A much more important consideration than the formation of damps, however, is the proper method of avoiding their practical effects. The inflammation lies at one kind develops on eye methods of preventing it from accumulating, viz., by letting fire to it. This may be done with flying under it has been offered to get too far before the very moment is made: Im the insan- hammable air, being much lighter than any other kind will naturally fit to the day so that a man, ly- ing flat on the ground to avoid the force of the ex- plofion, and holding up a lighted candle fixed upon a pole, you at once fee the mine from fetch a trouble- home goodt. But where it has been allowed to ac- cumulise in too great quantity, fo that this method causes be ufeid, or in the other kind, which is not in- flannable, the method commonly justified is to pro- duce a confluent circulation of air as much as possible through all parts of the mine. To procure this they make a perpendicular opening, which they call _haunk_ or _flaqls_, to that the mine may have two or more openings; and thus by realtion of the difference of em- perature between the open atmosphere and that in the mine, there is a constant danger of air through them both. This current will always be fronger in proportion to the difference between the external at- mendapre and that of the mine; and likewise in pro- portion to the difference between the depth of the two thafts. But as the temperature of the atmosphere is variable, it happens, at certain feafons of the year, that there is not a sufficient difference between that DAM of the atmosphere and in the mine to produce the ne- ceffiny circulation. This happens principally in the firing

and assembly, at which relation if necessary to fight iron in the fluids, which are always efficacious for the purpose defend.

Among the other afes to which dephloglitated air, might be applied, Mr Cavallo reckona that of recurring people from the dangerous effects of dams in mines, and some fishermenous places. If a large blad- der;" fays he, " into which a foliation of lime in va- ter is introduced, be filled with dephlogilitated air, and in his wooden or glala pipe. He adapted to its deck, a man may hold that pipe in his mouth, and may breathe the ofphiglicated clay, and thus equip- pedel be may enter into the fubtermous places, until the various shade fluids contained in them. A large bladder of dephiglittered air will fever be above a quarter of an hour, which is a length of time follicles for various purposes besides, it longer time it required to be feet in these places, aper- do may have two or more bladder of dephiglitated air along with him, and may fihlit as foot as the air of use is contaminated. Without the receiptify of any more complicated approach, the diabetes full of dephloglicitated air may be kept flopped by putting corks into the gills or wooden pipes that are tied to their necks. This is mainly allo to feed the diving balls."

Source record

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Damps. thickened the air in the pit, and then it would appear again; they lighted their candles at it ſometimes when they went out; and ſo in this pit it did no further harm."

In another pit, however, it ſoon appeared, and at laſt produced a moſt terrible exploſion. This was occaſioned by one of the workmen going imprudently down with a lighted candle, after a ceſſation of work for ſome days, and the force exerted by it ſeemed equal to that of gun-powder.

The formation of inflammable air in mines cannot be attributed to any vitiation of the atmoſphere; for there is no natural proceſs with which we are acquainted, by which ſuch a change could be accompliſhed. In one inſtance, however, we have an example of a fire-damp being produced, not only without any conſiderable ſtagnation of atmoſpherical air, but where there is the beſt circulation imaginable. This is in large bellows uſed in metallurgic works, which are ſometimes burſt by an exploſion of inflammable matter proceeding from the rancid matters with which the leather is greaſed. Dr Prieſtley has ſhown, that inflammable air is compoſed of pure elementary fire, charcoal or phlogiſton, and a little water; and that this compoſition may take place even *in vacuo*. All theſe materials are to be met with in the bowels of the earth. Coal, a bituminous ſubſtance, is abundantly able to ſupply the phlogiſton; the natural moiſture of the earth affords water, and the heat, however gentle, which conſtantly exiſts in the bowels of the earth, may be ſufficient to produce a quantity of inflammable air, which gradually accumulating in thoſe places where there is not a conſtant ſtream of atmoſpherical air to carry it off, will ſoon produce the dreadful effects already mentioned.

A much more important conſideration than the formation of damps, however, is the proper method of avoiding their pernicious effects. The inflammability of one kind affords an eaſy method of preventing it from accumulating, viz. by ſetting fire to it. This may be done with ſafety, unleſs it has been ſuffered to go too far before the experiment is made: for the inflammable air, being much lighter than any other kind, will naturally riſe to the top; ſo that a man, lying flat on the ground to avoid the force of the exploſion, and holding up a lighted candle fixed upon a pole, may at once free the mine from ſuch a troubleſome gueſt. But where it has been allowed to accumulate in too great quantity, ſo that this method cannot be uſed, or in the other kind, which is not inflammable, the method commonly practiſed is to produce a conſtant circulation of air as much as poſſible through all parts of the mine. To procure this, they make a perpendicular opening, which they call a *ſhank or ſhaft*, ſo that the mine may have two or more openings; and thus by reaſon of the difference of temperature between the open atmoſphere and that in the mine, there is a continual draught of air through them both. This current will always be ſtronger in proportion to the difference between the external atmoſphere and that of the mine; and likewiſe in proportion to the difference between the depth of the two ſhafts. But as the temperature of the atmoſphere is variable, it happens, at certain ſeaſons of the year, that there is not a ſufficient difference between that of the atmoſphere and in the mine to produce the neceſſary circulation. This happens principally in the ſpring and autumn; at which ſeaſons it is neceſſary to light fires in the ſhafts, which are always efficacious for the purpoſe deſired. Damſel ‖ Danae.

Among the other uſes to which dephlogiſticated air might be applied, Mr Cavallo reckons that of ſecuring people from the dangerous effects of damps in mines, and other ſubterraneous places. "If a large bladder," ſays he, "into which a ſolution of lime in water is introduced, be filled with dephlogiſticated air, and a ſmall wooden or glaſs pipe be adapted to its neck, a man may hold that pipe in his mouth, and may breathe the dephlogiſticated air; and thus equipped he may enter into theſe ſubterranean places, amidſt the various claſtic fluids contained in them. A large bladder of dephlogiſticated air will ſerve for above a quarter of an hour, which is a length of time ſufficient for various purpoſes; beſides, if longer time is required to be ſpent in theſe places, a perſon may have two or more bladders of dephlogiſticated air along with him, and may ſhift as ſoon as the air of one is contaminated. Without the neceſſity of any more complicated apparatus, the bladders full of dephlogiſticated air may be kept ſtopped by putting corks into the glaſs or wooden pipes that are tied to their necks. This air might alſo be uſed for diving-bells."

DAMSEL, from the French *damoiſel* or *damoiſeau*, an appellation anciently given to all young people of either ſex, that were of noble or genteel extraction, as the ſons and daughters of princes, knights, and barons: thus we read of Damſel Pepin, Damſel Louis le Gros, Damſel Richard prince of Wales.

From the ſons of kings this appellation firſt paſſed to thoſe of great lords and barons, and at length to thoſe of gentlemen who were not yet knights.

At preſent damſel is applied to all maids or girls not yet married, provided they be not of the vulgar.

DAN, or Jor-dan, which laſt literally denotes "the river Dan;" ſo named from the people where it has its ſource, which is a lake called *Phiala*, from its round figure, to the north of its apparent riſing from the mountain Panium or Paneum, as was diſcovered by Philip, Tetrarch of Trachonites; for on throwing light bodies into the Phiala, he found them to emerge again at Paneum (Joſephus). From Paneum it runs in a direct courſe to a lake called *Samachonites*; as far as which it is called *Jordan the Leſs*; and thence to the lake Geneſareth, or of Tiberias, where it comes increaſed by the lake Samachonitis and its ſprings, and is called the *Greater Jordan*; continuing its direct courſe ſouthwards, till it falls into the Aſphaltites.

Dan (anc. geog.), a town to the weſt of the ſource of the Jordan; formerly called *Laiſ* (Joſhua, Judges, Joſephus). This was the north, as Beerſheba was the ſouth, boundary of the Iſraelites; as appears from the common expreſſion in Scripture, from *Dan* to *Beerſheba*. At Dan Jeroboam erected one of the golden calves (1 Kings xii.).

Dan, the tribe, extended itſelf weſtward of Judah, and was terminated by Azotus and Dora on the Mediterranean (Joſephus).

DANAE, in antiquity, a coin ſomewhat more than an obolus, uſed to be put into the mouths of the dead, to pay their paſſage over the river Acheron.

Figure 1: *
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